

Theory 13.8.6

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1 Aggregation by attention

The aggregation methods discussed so far either weight the contribution of the neighbors equally or in a way that depends on the graph topology. Conversely, in graph attention layers, the weights depend on the data at the nodes. A linear transform is applied to the current node embeddings so that:

$$\mathbf{H}'_k = \beta_k \mathbf{1}^T + \Omega_k \mathbf{H}.$$

Then the similarity s_{mn} of each transformed node embedding $\mathbf{h}'_m$ to the transformed node embedding $\mathbf{h}'_n$ is computed by concatenating the pairs, taking a dot product with a column vector ϕ_k of learned parameters, and applying an activation function:

$$s_{mn} = a \left(\phi_k^T \begin{bmatrix} \mathbf{h}'_m \\ \mathbf{h}'_n \end{bmatrix} \right).$$

These variables are stored in a matrix $\mathbf{S}$ of size $N \times N$, where each element represents the similarity of every node to every other. As in dot-product self-attention, the attention weights contributing to each output embedding are normalized to be positive and to one using the softmax operation. However, only those values corresponding to the current node and its neighbors should contribute. The attention weights are applied to the transformed embeddings:

$$\mathbf{H}_{k+1} = \mathbf{a} \left[\mathbf{H}'_k \cdot \text{Softmask}[\mathbf{S}, \mathbf{A} + \mathbf{I}] \right]$$